TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

Analysis of Group V (Mg, Na, K, NH_4^+)

The soluble group — no group reagent and no precipitate: magnesium by its phosphate, the alkali metals by their flames, and ammonium, tested first of all, by its gas and Nessler's reagent.

CHE-1106 · CHEMISTRY-I SESSIONAL · EXPERIMENT 10 · LEVEL 1 TERM 1

1. Aim and the group's odd character

Aim: to detect Mg^{2+}, Na^+, K^+ and NH_4^+ — the cations that survive the whole separation scheme in the final filtrate. Group V has **no group reagent**: its members' compounds are too soluble for any common precipitant, so each is hunted individually. Two disciplines follow: the final filtrate of Experiment 4 is concentrated (evaporated down) before testing, since five steps of washing have diluted it; and NH_4^+, though a member, is proved on the **original salt** (Experiment 2's standing rule), never on a filtrate that has met NH_4Cl and NH_4OH.

2. Magnesium

TEST	DETAILS AND RESULT
Phosphate test (the confirmation)	To a portion of the concentrated filtrate add NH_4Cl, NH_4OH (to make it ammoniacal) and disodium hydrogen phosphate (Na_2HPO_4); scratch the inside of the tube with a glass rod and let it stand: a **white crystalline precipitate** of magnesium ammonium phosphate ($MgNH_4PO_4 \cdot 6H_2O$) confirms Mg^{2+}. Crystallinity and the need for standing/scratching distinguish it from amorphous white precipitates.
Supporting behaviour	NaOH gives white $Mg(OH)_2$, insoluble in excess (contrast Zn and Al — the negative that places Mg in this group at all).

If the filtrate carries large amounts of ammonium salts from the scheme, they are first expelled (evaporate and ignite gently, take up the residue in dilute HCl) before the phosphate test, or the precipitate forms reluctantly and impurely.

3. Sodium and potassium — the flame pair

CATION	FLAME	WET CONFIRMATION
Na^+	Intense, persistent golden yellow — so strong that traces (from glass, fingers, dust) fake it; report sodium only when the flame is strong and lasting	No convenient precipitate in this scheme; the flame, strongly positive, is the report. (Zinc uranyl acetate gives a yellow crystalline precipitate where the laboratory keeps it.)
K^+	Lilac/violet, brief and easily masked by sodium — view through **cobalt-blue glass**, which absorbs the sodium yellow; through the glass, potassium appears crimson-purple	Sodium cobaltinitrite added to the neutral/acetic solution gives a **yellow precipitate** (potassium sodium cobaltinitrite) on standing. Perchloric acid gives white $KClO_4$ where used.

4. Ammonium — tested first, confirmed twice

TEST	DETAILS AND RESULT
Alkali test	Warm the **original salt** with NaOH solution: ammonia gas evolves — recognised by its smell, by turning moist red litmus blue, and by dense white fumes with a glass rod dipped in concentrated HCl.
Nessler's test (the confirmation)	To a very dilute solution (or to the gas absorbed in a little water) add Nessler's reagent (alkaline K_2HgI_4): a **brown colour or brown precipitate** even at great dilution confirms NH_4^+.

Nessler's reagent contains mercury — its residues follow the heavy-metal waste rule of Experiment 5, and the test is done on drops, not beakers.

5. Order of work on the final filtrate

1. Evaporate the final filtrate to a small volume; if ammonium salts burden it, expel them by evaporation/ignition and redissolve as §2 directs.
2. Test one portion for Mg^{2+} (phosphate test).
3. Test the flame on a clean wire: sodium's blaze first; through cobalt glass for potassium; confirm potassium wet if reagents allow.
4. Report NH_4^+ only from the original-salt tests (alkali + Nessler), recorded at the head of the analysis where they were performed.

6. Practice (viva preparation)

Answer in your own words; brief model answers follow.

1. Why has Group V no group reagent, and what two working consequences follow?
2. Give the phosphate test for Mg^{2+}: reagents, conditions, the precipitate's name and its two identifying manners.
3. A flame is flooded golden yellow. Which cation does it prove, what does it hide, and through what is the hidden one sought?
4. State the wet confirmation for K^+ with sodium cobaltinitrite.
5. Why must NH_4^+ be absent (or expelled) before a clean magnesium phosphate test, and why is NH_4^+ itself never proved on the final filtrate?
6. Give both ammonium tests and the waste rule that follows one of them.

Answer key

1. Because all common compounds of Mg^{2+}, Na^+, K^+ and NH_4^+ are too soluble for any reagent to precipitate the four together. Consequences: each member is tested individually, and the final filtrate is concentrated first because the scheme's washings have diluted it.
2. In ammoniacal solution (NH_4Cl + NH_4OH present) add Na_2HPO_4; scratch and stand: white crystalline magnesium ammonium phosphate falls. Manners: it is crystalline (not gelatinous/amorphous) and forms slowly, encouraged by scratching.
3. It proves Na^+ (when strong and persistent). It hides K^+'s lilac; potassium is sought through cobalt-blue glass, which absorbs the yellow and passes potassium's crimson-purple.
4. Sodium cobaltinitrite in neutral/slightly acetic solution gives a yellow precipitate of potassium sodium cobaltinitrite on standing.
5. Excess ammonium salts keep magnesium in solution and make the phosphate precipitate slow and impure, so they are expelled first. And the filtrate has been treated with NH_4Cl/NH_4OH throughout the scheme — any ammonium "found" there could be reagent, not sample; NH_4^+ is proved only on the original salt.
6. Warm with NaOH → ammonia (smell, red litmus blue, white fumes with HCl rod); Nessler's reagent → brown colour/precipitate. Nessler's contains mercury: residues to heavy-metal waste.
